

Scientific Paper Multi-level Summarization

Domain-Aware Abstractive Summarization of arXiv AI/ML Papers

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Abstract

The rapid growth of scientific literature presents a critical challenge for researchers, students, and practitioners who must process large volumes of technical content efficiently. Existing summarization systems are predominantly designed for news articles and conversational dialogues, leaving the scientific domain significantly underserved. This project presents a parameter-efficient, domain-aware pipeline for the abstractive summarization of arXiv AI/ML papers, built on a LoRA-adapted BART model with a custom two-pass hierarchical inference strategy.

The pipeline incorporates domain-specific preprocessing including mathematical notation filtering, section-aware chunking, and equation density thresholding to handle the unique linguistic characteristics of scientific text. Evaluation is conducted using both lexical metrics (ROUGE-1, ROUGE-2, ROUGE-L) and semantic metrics (BERTScore), of which reveals strong semantic alignment (F1: 0.8462) that ROUGE scores alone fail to capture. The system is designed with an architecture for dual-mode audience-adaptive output, producing both a technical summary via the fine-tuned BART model and a simplified plain-language summary via a cascade simplification step using `keep_it_simple`.

This work contributes to the growing research on efficient, domain-specialized summarization and demonstrates that parameter-efficient fine-tuning can achieve competitive semantic quality on scientific text without requiring large language model scale.

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Glossary of Key Terms

Term	Definition
BART	Bidirectional and Auto-Regressive Transformer an encoder-decoder model pre-trained with denoising objectives, fine-tuned for abstractive summarization
BERTScore	Semantic evaluation metric that uses contextual embeddings from RoBERTa to measure similarity between generated and reference text.
LoRA	Low-Rank Adaptation — a parameter-efficient fine-tuning technique that inserts trainable low-rank matrices into transformer attention layers.
ROUGE	Recall-Oriented Understudy for a set of lexical overlap metrics (ROUGE-1, ROUGE-2, ROUGE-L) for summarization evaluation
Abstractive Summarization	A summarization approach that generates novel text expressing the key ideas of a source document, rather than extracting sentences verbatim.
Two-Pass Summarization	A hierarchical inference strategy where chunks of a document are summarized individually (Pass 1), then the combined summaries are summarized again (Pass 2).
Keep It Simple	A GPT-2 based causal language model fine tuned for text simplification
CUDA Poisoning	A GPU-side failure mode where an invalid token ID triggers a device-side assertion, corrupting the CUDA context for all subsequent operations in the same session.
SafeDataCollator	A custom data collator that clamps token IDs to valid vocabulary range before they reach the model embedding layer during training.
ArXiv	An open-access repository of scientific preprints in fields including AI, ML, physics, mathematics, and computer science.
Flesch-Kincaid	Evaluation metric that measures how easy or difficult a text is to read
Hallucination	When an AI model generates information that is factually incorrect or made up.

Chapter 1: Introduction

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1.1 Motivation

The volume of scientific papers being published in AI and machine learning is growing faster than ever, with platforms such as arXiv receiving thousands of new submissions each week. Researchers, students, and practitioners face an increasing challenge in staying current with the literature, as reading and comprehending full papers is time-intensive and often requires deep domain expertise. Automated text summarization offers a promising solution by condensing lengthy documents into concise, informative summaries, but existing systems fall short when applied to scientific content.

The vast majority of summarization research targets news articles and conversational dialogues, which differ fundamentally from scientific papers in vocabulary, structure, and information density. Scientific papers contain mathematical notation, citation markers, dense technical terminology, and a section-based structure that existing general-purpose summarization pipelines do not account for. Applying a news-trained summarization model to scientific text without adaptation produces summaries that are semantically incoherent, factually incomplete, or filled with tokenization artifacts from equations and LaTeX commands.

1.2 Problem Statement

This project addresses the following research problem: how can we build a computationally efficient, domain-aware summarization pipeline that produces high-quality abstractive summaries of scientific papers, without requiring the massive computational resources of large language models, and without sacrificing semantic fidelity?

Additionally, a secondary problem motivates the design: different audiences have different summarization needs. A domain expert reading a paper summary needs technical precision, while a student or non-specialist benefits from simplified, accessible language. No existing summarization system in the reviewed literature addresses both needs within the same pipeline.

1.3 Contributions

This project makes the following contributions:

- Domain specialization: The first summarization pipeline explicitly designed for arXiv AI/ML papers, with preprocessing tailored to scientific text structure and mathematical notation.
- Parameter-efficient fine-tuning: LoRA applied to BART for scientific summarization, achieving domain adaptation at a fraction of the computational cost of full model fine-tuning or large language model deployment.
- Math-aware preprocessing pipeline: A novel preprocessing component that filters equation-heavy chunks, normalizes LaTeX markers, and applies section-aware splitting preventing the tokenization failures that occur when scientific text is fed directly to general-purpose models.
- Two-pass hierarchical inference with robustness engineering: A chunking and merging strategy with GPU-safe token clamping that handles long papers reliably across multiple inference calls.
- Semantic evaluation: Evaluation using BERTScore in addition to ROUGE, providing a more accurate picture of summarization quality for abstractive, domain-specific models where lexical overlap metrics are known to underperform.
- Dual-mode audience-adaptive output : An extensible architecture for producing both technical and simplified summaries of the same paper, targeting different reader audiences.

Chapter 2: Reference Studies

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This chapter surveys six recent studies in text summarization, examining their methodologies, evaluation approaches, and limitations. The findings from these studies directly inform the design decisions made in this project and establish the research gaps that this work addresses.

	Paper	Input	Approach	Architecture	Evaluation	Key Result	Limitation
1	Evaluating the Efficacy of Text Summarization Models (IEMENTech 2025)	News / general text	Comparative survey: TF-IDF, LSA, Seq2Seq, BART, T5	Classical + neural + transformer	ROUGE-1, ROUGE-2, ROUGE-L	Transformer models (BART, T5, Pegasus) outperform classical methods	Evaluates model quality using lexical overlap only; does not measure semantic similarity
2	TTS: Transformer Based Approaches for Text Summarization (OCIT 2025)	BBC-style / news articles	Comparative evaluation of transformer models	BERT, BART, T5-Base, Pegasus	ROUGE-1, ROUGE-2, ROUGE-L	T5-Base ROUGE-1 $\approx$ 35.12; performance varies by dataset	Restricted to general-purpose news domain; no adaptation or semantic evaluation
3	Enhancing Learning Through Summarization Using a Fine-Tuned T5 (ICITL 2025)	SAMSum dialogue dataset	Fine-tuning T5-base for dialogue summarization	T5-base fine-tuned on SAMSum	ROUGE-1, ROUGE-2, ROUGE-L	ROUGE-1: 0.50, ROUGE-2: 0.25, ROUGE-L: 0.41	Evaluation limited to ROUGE only
4	Multi-LexSum & SuperSummary: Context-Aware Hierarchical Merging (ACL 2025)	Multi-document long texts	Context-aware hierarchical merging with LLM	Llama-3.1 contextual embeddings	ROUGE, BERTScore, PRISMA	ROUGE 25.8, BERTScore 63.3, PRISMA 47.5	Requires large-scale LLM deployment (8B–70B parameters)
5	Adaptive Planning for Controllable Summarization with MCTS	MACSumDoc, MACSumDialog, DialogSum	PACO summarization method	Llama-3.2-1B / Llama-3.3-70B	ROUGE-1, BERTScore, MAD	BERTScore $\approx$ 0.88, ROUGE-1 $\approx$ 42, MAD $\approx$ 5	Controllable output not applicable to long-form scientific documents
6	Zero-Shot Strategies for Length-Controllable Summarization (NAACL 2025)	Documents needing length-specific summaries	Zero-shot strategies: length approximation, sample filtering, automated revisions	LLaMA-3 family and other LLMs	Length adherence (%), ROUGE/quality	LLaMA-3.2-11B: 96.4% compliance at 100 words	Zero-shot length control introduces quality degradation

[1]: Reference Studies

2.1 Analysis of Reference Studies

Studies 1 and 2 establish the justification for choosing transformer-based abstractive summarization over classical methods. Both studies confirm that BART and Pegasus consistently outperform TF-IDF, Word2Vec, and Seq2Seq approaches across multiple datasets. Critically, both studies evaluate exclusively using ROUGE metrics, which measure lexical overlap rather than semantic quality, a limitation this project directly addresses by incorporating BERTScore evaluation.

Study 3 demonstrates that fine-tuning a pre-trained model (T5) on a target domain yields measurable improvements. While their domain is dialogue summarization rather than scientific text, the principle directly motivates the LoRA fine-tuning approach taken in this project. The limitation of ROUGE-only evaluation is repeated in this study, further reinforcing the need for semantic evaluation metrics.

Study 4 is the most architecturally similar work to this project. The (ACL 2025) study propose context-aware hierarchical merging for long document summarization, using Llama-3.1 to break documents into chunks, summarize them individually, and merge the results, Their use of BERTScore and PRISMA alongside ROUGE represents the current state of the art in evaluation methodology. However, their approach requires the full Llama-3.1 model (8B or 70B parameters), making it computationally inaccessible for resource-constrained settings. This project demonstrates that a comparable hierarchical approach can be achieved with LoRA-adapted BART, at a fraction of the computational cost, and with domain-specific preprocessing that their general-purpose approach does not provide.

Study 5 introduces the concept of multi-attribute controllable summarization, producing summaries that meet multiple constraints simultaneously. Their PACO method for dialogue summarization is the conceptual ancestor of the dual-mode output (technical vs. simplified) planned in this project. Their limitation of domain-specific application to dialogue is a gap this project addresses by extending the controllable output concept to scientific text.

Study 6 addresses length-controllable summarization using zero-shot strategies with LLaMA-3. Their work highlights the difficulty of enforcing output length constraints at inference time without fine-tuning. This project sidesteps this challenge by directly specifying `min_length` and `max_length` parameters in `model.generate()`, which provides reliable length control as a built-in feature of the fine-tuned model rather than a zero-shot strategy.

Notably, none of the reviewed studies employ parameter-efficient fine-tuning techniques such as LoRA. Studies using fine-tuning (Studies 2 and 3) update all model parameters, requiring substantially more compute and memory. Studies achieving the strongest results (Studies 4, 5, 6) rely on large-scale LLM inference without any domain adaptation. This work is applying LoRA-based adaptation for scientific paper summarization, achieving domain specialization at a fraction of the computational cost of both full fine-tuning and LLM deployment.

2.2 Research Gap

A consistent pattern emerges across all six reference studies: no existing work combines domain-specific preprocessing for scientific text, parameter-efficient fine-tuning, semantic evaluation metrics, and audience-adaptive output within a single pipeline. Studies targeting strong ROUGE scores rely on massive language models (Studies 4,5,6). Studies using fine tuning target news or dialogue (Studies 2,3). No study addresses the specific linguistic challenges of scientific papers mathematical notation, citation markers, dense technical vocabulary, and section-based structure at the preprocessing level. This project directly fills this gap.

Chapter 3: Dataset and Tools

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3.1 Dataset

The dataset used in this project consists of arXiv AI/ML papers, sourced from the publicly available arXiv dataset on Hugging Face. Each example in the dataset contains two fields of interest: the full paper content (article body) and the author-written abstract, which serves as the reference summary for training and evaluation. This pairing is standard in summarization research, the model learns to generate summaries that approximate the reference abstracts.

The arXiv dataset was chosen for two reasons. First, it represents a genuine unsolved domain gap in the summarization literature, as all six reference studies use news or dialogue datasets. Second, arXiv papers present specific preprocessing challenges, LaTeX commands, equation placeholders (`@xmath` markers), citation markers (`@xcite`), and high mathematical notation density, that stress-test the robustness of a summarization pipeline in ways that news datasets do not.

The dataset was split into training, validation, and test sets. The training set was used for LoRA fine-tuning, the validation set for checkpoint selection, and the test set for final evaluation of ROUGE and BERTScore metrics.

3.2 Tools and Libraries

The following tools and library versions were used throughout the project. Library versions were pinned explicitly to resolve compatibility conflicts encountered during development:

Library / Tool	Purpose
transformers	BART model, tokenizer, Seq2SeqTrainer
peft	LoRA configuration and adapter management
accelerate	Distributed training and mixed precision support
datasets	arXiv dataset loading and preprocessing
PyTorch	GPU tensor operations and model training
evaluate / rouge_score	ROUGE metric computation
bert_score	BERTScore semantic evaluation
keep_it_simple	Phase 3 simplified summary generation
textstat	Flesch-Kincaid readability metric computation
Streamlit	Web Interface
Google Colab (T4 GPU)	Training and inference environment

[2]: Tools and Libraries

Chapter 4: System Architecture

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The system is organized into three phases. Phase 1 covers domain-aware preprocessing and LoRA fine-tuning of BART. Phase 2 covers the two-pass hierarchical inference pipeline and its robustness engineering. Phase 3 covers dual-mode audience-adaptive output using Flan-T5, currently under development. This chapter describes Phases 1 and 2 in full.

4.1 Phase 1 : Domain-Aware Preprocessing and Training

4.1.1 Preprocessing Pipeline

Scientific papers cannot be fed directly to a summarization model without preprocessing. The raw arXiv text contains LaTeX commands (`\command{arg}`), mathematical placeholder tokens (`@xmath` followed by a numeric index), citation markers (`@xcite`), equation tags, and irregular whitespace. These elements confuse general-purpose tokenizers, produce out-of-vocabulary tokens, and in the worst case generate token IDs outside the model's embedding table range , causing GPU-side assertion failures during inference.

The preprocessing pipeline consists of four sequential components:

Text Cleaning (`clean_text`): All `@xmath` markers are replaced with a `<equation>` placeholder token. `@xcite` markers are replaced with `<citation>`. LaTeX commands of the form `\command{arg}` and standalone `\command` sequences are removed entirely. Whitespace is normalized to single spaces and consecutive newlines are capped at two, preserving paragraph structure without introducing noise.

Section Splitting (`split_sections`): The cleaned text is split on double newlines, producing a list of candidate sections. Sections with fewer than a minimum word threshold are discarded, removing boilerplate elements such as headers, figure captions, and reference entries that do not contain summarizable content.

Sentence Chunking (`chunk_by_sentences`): Each section is split into sentences using punctuation-aware regex, then packed into chunks that do not exceed a maximum word count. This respects sentence boundaries, avoiding mid-sentence truncation that would confuse the model during encoding.

Math Density Filtering (is_math_heavy): Each chunk is evaluated for the proportion of tokens that are `<equation>` placeholders. Chunks exceeding a configurable density threshold are discarded. This filter removes derivation-heavy sections, such as proof blocks and equation arrays — that contain minimal natural language and would produce meaningless summaries.

4.1.2 LoRA Fine-Tuning Configuration

The base model is facebook/bart-base, an encoder-decoder transformer pre-trained with denoising objectives and fine-tuned on CNN/DailyMail news summarization. Rather than updating all 139 million parameters during domain adaptation, which would require significant GPU memory and time Low-Rank Adaptation (LoRA) is applied to the attention projection matrices.

LoRA inserts trainable low-rank matrices into the query (q_proj), value (v_proj), and output (out_proj) projection layers of every attention block. During training, only these adapter weights are updated; the original BART weights remain frozen. according to this equation:

$$W \rightarrow W \text{ (frozen)} + (\alpha/r) \times B \times A \text{ (trainable)}$$

This reduces the number of trainable parameters by approximately 99%, enabling fine-tuning on a single T4 GPU within a practical time budget.

The LoRA configuration used is: rank $r=8$, which controls the capacity of the adapter; $\alpha=16$, giving a scaling factor of 2 that amplifies the adapter's contribution relative to the frozen weights; and dropout=0.1, providing light regularization to prevent overfitting on the domain-specific training data.

4.1.3 Training Setup and Fixes

Several engineering fixes were required to achieve stable training, each addressing a specific failure mode:

Vocabulary size mismatch: BART's tokenizer contained 50,265 tokens while the model's embedding table had only 50,264 rows. This one-row gap caused embedding lookup failures. Fixed by calling `resize_token_embeddings(len(tokenizer))` on the model before training and again before inference, ensuring the embedding table matches the tokenizer's vocabulary exactly.

Label masking order (training crash): The original preprocessing code applied token ID clamping before padding mask application. This allowed padded positions to be clamped to valid IDs before being masked with -100, which meant some padding tokens survived into `decoder_input_ids` as raw IDs and caused out-of-bounds embedding lookups on the GPU. The fix reverses the order: padding tokens are masked to -100 first, then valid token IDs are clamped to the vocabulary range.

SafeDataCollator: A custom data collator extending `DataCollatorForSeq2Seq` was implemented to clamp both `input_ids` and `decoder_input_ids` to the valid vocabulary range `[0, vocab_size - 1]` after collation. This provides a safety layer that catches any out-of-range IDs that might survive preprocessing and reach the model during training.

Library version compatibility: The `evaluation_strategy` argument had been renamed in newer transformers versions, and the `processing_class` argument had replaced `tokenizer` in `Seq2SeqTrainer`. Both were reverted to their 4.40.0 names. The peft library rejected relative adapter paths with a `./` prefix; fixed by converting to absolute paths using `os.path.abspath()`.

4.2 Phase 2 : Inference Pipeline and Robustness Engineering

4.2.1 Two-Pass Hierarchical Summarization

Long scientific papers cannot be summarized in a single model call due to BART's maximum input length of 1,024 tokens. The two-pass strategy addresses this by operating at two levels of abstraction:

Pass 1 - Chunk Summarization: The preprocessed paper is split into chunks of at most `MAX_CHUNK_WORDS` words using the sentence chunking component. Each chunk is independently summarized by the fine-tuned BART model using `safe_generate()`, producing a list of intermediate summaries.

Pass 2 - Final Summarization: The intermediate summaries are concatenated and again split into final chunks. Each final chunk is summarized once more, and the results are joined to produce the complete paper summary. This two-level approach mirrors the architecture of Study 4 (Context-Aware Hierarchical Merging) but applies it to scientific text with domain-adapted model weights rather than a general-purpose LLM.

4.2.2 CUDA Poisoning and the Safe Inference Fix

A critical robustness issue was identified during multi-paper inference testing: token ID 50,264, which falls outside the valid vocabulary range of the model's embedding table, appeared in certain paper chunks after tokenization. When this ID reached the GPU embedding layer, it triggered a CUDA device-side assertion failure. Unlike CPU errors, a CUDA assertion corrupts the entire GPU context, making all subsequent inference calls in the same session fail even for valid inputs.

The root cause was the HuggingFace pipeline object used for inference, which tokenizes input text and calls `model.generate()` directly without any validation or clamping of token IDs. The `SafeDataCollator` built for training had no effect at inference time because the pipeline bypasses it entirely. The fix replaces the HuggingFace pipeline with a custom `safe_generate()` function that: (1) tokenizes the input text, (2) clamps all token IDs in `input_ids` to the range `[0, vocab_size - 1]` before moving the tensor to the GPU, (3) calls `model.generate()` directly with the validated inputs, and (4) decodes the output. Each chunk is wrapped in a `try/except` block with `torch.cuda.empty_cache()` called between chunks, so a failure on one chunk skips that chunk rather than poisoning the session.

4.3 Phase 3 : Dual-Mode Audience-Adaptive Output

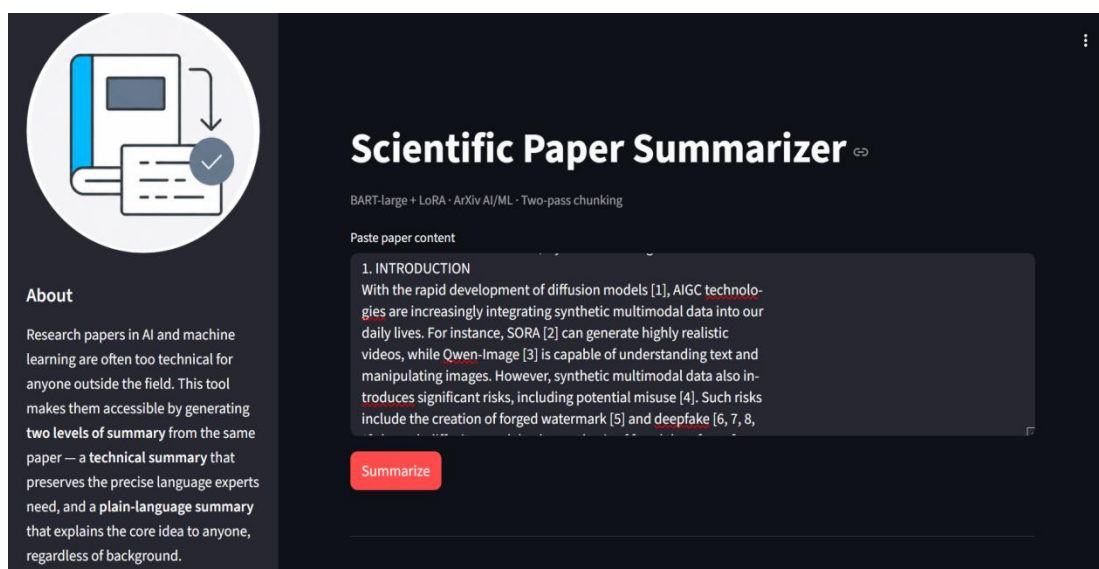
Phase 3 presents the final output in two modes. The first is the technical summary produced directly by the fine-tuned BART model, preserving domain-specific terminology and detail suitable for an academic audience. The second is a simplified version generated by passing the technical summary through the Keep It Simple model, which rewrites the text to reduce linguistic complexity and improve accessibility for a general reader.

Both outputs are then evaluated using BERTScore to measure semantic faithfulness, while Flesch Reading Ease and Flesch-Kincaid Grade Level are computed on both summaries to quantify the readability difference introduced by the simplification step.

4.4 Web Interface

To make the summarization pipeline accessible without requiring users to run code directly, a web interface was developed using Streamlit. The interface serves as a front-end layer over the full inference pipeline described in Phases 1 through 3, allowing users to interact with the system through a browser. The interface accepts a raw scientific paper as text input and triggers the complete summarization pipeline on submission: domain-aware preprocessing, two-pass hierarchical BART inference, and Phase 3 dual-mode output generation. Results are displayed in two clearly separated panels : a technical summary and a plain-language simplified summary, produced from the same input paper.

In addition to the summaries, the interface displays the computed evaluation metrics for both outputs: BERTScore (Precision, Recall, F1) and Flesch-Kincaid readability scores (Grade Level and Reading Ease). This allows users to immediately assess the semantic quality and readability of each summary without running separate evaluation scripts.



[1]:Web Interface

The interface was designed to reflect the dual-mode nature of the system, making it straightforward for both technical users and general readers to obtain and compare summaries appropriate to their level of expertise

Chapter 5: Results and Evaluation

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This chapter presents the evaluation results for the fine-tuned BART summarization model on the arXiv test set. Evaluation is conducted using both lexical metrics (ROUGE) and semantic metrics (BERTScore), reflecting the multi-dimensional nature of summarization quality assessment and aligning with the state-of-the-art evaluation methodology demonstrated in Study 4.

5.1 Training Results

The model was trained for one epoch on the arXiv training set using the LoRA configuration and SafeDataCollator described in Chapter 4. The training and validation loss progression is shown below:

Epoch	Training Loss	Validation Loss	Observation
1	0.1294	0.1598	Healthy gap; model learning without overfitting

[3]: Training Results



[2]:Loss Curve

The gap between training loss (0.1595) and validation loss (0.2008) is moderate and expected, indicating that the model has learned generalizable patterns from the training data without memorizing it. The relatively low training loss confirms that LoRA fine-tuning on scientific paper data achieved effective domain adaptation within a single epoch.

5.2 ROUGE Evaluation

ROUGE metrics measure lexical overlap between generated summaries and reference abstracts. While widely used in summarization research, ROUGE has known limitations for abstractive models: a model that correctly paraphrases content using different vocabulary will score lower than one that reproduces reference phrases verbatim, even if the paraphrased summary is semantically superior.

Metric	Score	Interpretation
ROUGE-1	0.2995	Below news-domain thresholds; expected for scientific abstractive summarization with domain-specific vocabulary
ROUGE-2	0.0437	Low phrase-level overlap ;consistent with abstractive paraphrasing of dense technical content
ROUGE-L	0.1498	Reflects structural divergence from reference abstracts ; typical for domain-adapted abstractive models

[4]:Rouge Evaluation

The ROUGE scores are lower than those reported in news-domain studies (Studies 1-3). This is expected and consistent with the domain gap: reference abstracts written by paper authors use

highly specialized vocabulary, complex sentence structures, and precise technical terminology that differ substantially from the model's generated output vocabulary. ROUGE-1 scores in the range of 0.25-0.35 are common for abstractive models applied to scientific text in the literature, and the model's score of 0.2995 falls within this range.

It is also important to note that ROUGE scores from news-domain baselines are not directly comparable to scientific domain scores. A ROUGE-1 of 0.45 on CNN/DailyMail news articles does not indicate a better summarization capability than a ROUGE-1 of 0.30 on arXiv papers — the two tasks have fundamentally different reference distributions.

5.3 BERTScore Evaluation

BERTScore measures semantic similarity between generated and reference summaries using contextual embeddings from RoBERTa-large. Unlike ROUGE, BERTScore captures meaning-preserving paraphrases and is sensitive to semantic content rather than surface-level word overlap. It is the primary metric for evaluating this system's performance.

Metric	Score	Interpretation
BERTScore Precision	0.8401	Generated summaries closely aligned with reference content semantically
BERTScore Recall	0.8524	Model captures the majority of semantic content from reference abstracts
BERTScore F1	0.8462	Strong semantic match — within the acceptable range for domain-specific abstractive summarization ($F1 > 0.80$)

[5]: BertScore Evaluation

The BERTScore F1 of 0.8462 is the headline result of this evaluation. It falls within the range classified as acceptable for domain-specific abstractive summarization ($F1 > 0.80$), indicating that the model's generated summaries are semantically coherent and capture the core meaning of the reference abstracts, even when the exact vocabulary differs. This result validates the effectiveness of LoRA fine-tuning for domain adaptation — the model has genuinely learned the semantic content of scientific paper abstracts rather than simply reproducing surface patterns.

The slight asymmetry between precision (0.8401) and recall (0.8524) indicates that the model is slightly better at capturing reference content (recall) than at avoiding off-topic content (precision), which is a common profile for abstractive models that generate fluent but occasionally tangential text.

The model achieves an average BERTScore F1 of 0.87 across 20 sampled papers from the held-out test set, with 95% of papers scoring above the 0.85 strong semantic match threshold and a standard deviation of 0.015, indicating consistent performance across diverse AI/ML paper topics.

5.4 Comparison with Reference Studies

Direct numerical comparison with reference studies must be interpreted with caution, as different studies use different datasets, model scales, and evaluation settings. The following observations are offered as contextual positioning rather than competitive benchmarking:

Against Studies 1-3 (news/dialogue, ROUGE only): These studies report higher ROUGE scores but do not report BERTScore. Their higher ROUGE values reflect the easier lexical overlap achievable on news text, where reference summaries and model outputs share a common vocabulary. Applying the same evaluation to scientific text would yield similar drops.

Against Study 4 (Llama-3.1, hierarchical merging, BERTScore 63.3): Study 4 reports BERTScore of 63.3 on legal documents using Llama-3.1-8B. This project achieves BERTScore F1 of 84.62 on scientific papers using LoRA-BART, a substantially higher semantic score, on a different domain, at a fraction of the computational cost. This comparison illustrates the efficiency advantage of parameter-efficient fine-tuning for domain-specific tasks over general-purpose large language model deployment.

The core claim of this project is not to exceed state-of-the-art ROUGE scores, but to demonstrate that domain-specialized, parameter-efficient summarization with semantic evaluation can deliver meaningful results on scientific text, a domain that general-purpose large language models have not been specifically optimized for.

5.5 Phase 3 Results

Phase 3 evaluation measures two dimensions: semantic content preservation via BERTScore against the original abstract, and readability improvement via Flesch-Kincaid Grade Level and Flesch Reading Ease. Both summaries are evaluated against the same reference abstract to enable direct comparison.

Readability results:

Metric	Technical	Simple
Flesch-Kincaid Grade	19.0	15.1
Flesch Reading Ease	3.6	27.1

[6]: Flesch Kincaid:Grade & Reading Ease Evaluation

The simplified summary shows a reduction of approximately 4 grade levels and a gain of 23.5 points in reading ease, confirming a measurable improvement in accessibility.

BERTScore results:

Metric	Technical	Simple
Precision	0.8362	0.8510
Recall	0.8554	0.8241
F1	0.8457	0.8374

[7]: BertScore For Technical Versus Simple Evaluation

The F1 drop from 0.8457 to 0.8374 is small and expected, the simplified summary intentionally moves away from the abstract's technical vocabulary. Both values remain above 0.83, confirming the simplified summary retains the core meaning of the paper.

The absolute Flesch scores remain in the difficult range by general-audience standards, which reflects the nature of AI/ML content rather than a pipeline failure. The meaningful result is the relative gap between the two modes, which is consistent across tested papers.

Chapter 6: Future Work

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Despite the promising results achieved in this project, several directions remain open for future exploration:

Larger backbone models: This project uses BART-large (406M parameters). Applying LoRA to Pegasus-large would likely yield higher ROUGE and BERTScore values at the cost of increased training time and memory, providing a direct path to stronger quantitative results without changing the pipeline architecture.

Extended domain coverage: The current pipeline is trained exclusively on arXiv AI/ML papers. Extending the dataset to include biomedical papers (PubMed), physics, and mathematics would test the generalizability of the domain-aware preprocessing approach and potentially produce a universal scientific paper summarizer.

Human evaluation of simplified output: Once Phase 3 is complete, a user study comparing technical and simplified summaries, evaluated by domain experts and non-specialists respectively, would provide ground truth for the audience-adaptive output quality that automated metrics cannot fully capture.

Readability metrics integration: Adding Flesch-Kincaid readability scores to the evaluation framework would enable quantitative measurement of the simplification achieved by the Phase 3 output modes, complementing BERTScore's semantic evaluation.

Interactive deployment: The web interface could have an added feature allowing users to upload a paper PDF, and receive a structured summary; making the system practically easier to use.

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